

1. NAME OF THE MEDICINAL PRODUCT

Pagenax 120 mg/ml solution for injection

2. QUALITATIVE AND QUANTITATIVE COMPOSITION

One ml solution for injection contains 120 mg of brolocizumab*.

* Brolocizumab is a humanised monoclonal single-chain Fv (scFv) antibody fragment produced in *Escherichia coli* cells by recombinant DNA technology.

Pagenax 120 mg/ml solution for injection

Each vial contains 27.6 mg brolocizumab in 0.23 ml solution. This provides a usable amount to deliver a single dose of 0.05 ml solution containing 6 mg of brolocizumab.

Excipient with known effect

Each vial contains 0.05 mg polysorbate 80 in 0.23 ml solution. This corresponds to 0.01 mg polysorbate 80 per dose (0.05 ml).

For the full list of excipients, see section 6.1.

3. PHARMACEUTICAL FORM

Solution for injection (injection).

Clear to slightly opalescent, colourless to slightly brownish-yellow aqueous solution.

4. CLINICAL PARTICULARS

4.1 Therapeutic indications

Pagenax is indicated in adults for the treatment of

- neovascular (wet) age-related macular degeneration (AMD) (see section 5.1),
- visual impairment due to diabetic macular oedema (DME) (see section 5.1).

4.2 Posology and method of administration

Pagenax must be administered by a qualified ophthalmologist experienced in intravitreal injections.

Posology

Wet AMD

Treatment initiation – loading

The recommended dose is 6 mg brolocizumab (0.05 ml solution), administered by intravitreal injection every 4 weeks (monthly) for the first 3 doses. A disease activity assessment is suggested 16 weeks (4 months) after treatment start.

Alternatively, 6 mg brolocizumab (0.05 ml solution) may be administered every 6 weeks for the first 2 doses. A disease activity assessment is suggested 12 weeks (3 months) after treatment start. A third dose may be administered based on disease activity as assessed by visual acuity and/or anatomical parameters at week 12.

Maintenance treatment

After the last loading dose, the physician may individualise treatment intervals based on disease activity as assessed by visual acuity and/or anatomical parameters. In patients without disease activity, treatment every 12 weeks (3 months) should be considered. In patients with disease activity, treatment every 8 weeks (2 months) should be considered. If patients are being treated according to a treat-and-extend regimen and there are no signs of disease activity, the treatment intervals could be extended stepwise until signs of disease activity recur. The treatment interval should be extended or shortened by no more than 4 weeks (1 month) at a time (see section 5.1). There are limited data on treatment intervals longer than 20 weeks (5 months). The treatment interval between two doses of Pagenax should not be less than every 8 weeks (2 months) (see section 4.4).

If visual and anatomical outcomes indicate that the patient is not benefiting from continued treatment, Pagenax should be discontinued.

DME

The recommended dose is 6 mg brolucizumab (0.05 ml solution) administered by intravitreal injection every 6 weeks for the first 5 doses.

Thereafter, the physician may individualise treatment intervals based on disease activity as assessed by visual acuity and/or anatomical parameters. In patients without disease activity, treatment every 12 weeks (3 months) should be considered. In patients with disease activity, treatment every 8 weeks (2 months) should be considered. After 12 months of treatment, in patients without disease activity, treatment intervals up to 16 weeks (4 months) could be considered (see sections 4.4 and 5.1).

If visual and anatomical outcomes indicate that the patient is not benefiting from continued treatment, Pagenax should be discontinued.

Special populations

Elderly

No dose adjustment is required in patients aged 65 years or above (see section 5.2).

Renal impairment

No dose adjustment is required in patients with renal impairment (see section 5.2).

Hepatic impairment

Brolucizumab has not been studied in patients with hepatic impairment. No dose adjustment is required in patients with hepatic impairment (see section 5.2).

Paediatric population

The safety and efficacy of brolucizumab in children and adolescents below 18 years of age have not been established. No data are available.

Method of administration

Pagenax is for intravitreal use only.

The solution for injection should be inspected visually prior to administration (see section 6.6).

The intravitreal injection procedure should be carried out under aseptic conditions, which includes the use of surgical hand disinfection, sterile gloves, a sterile drape and a sterile eyelid speculum (or equivalent). Sterile paracentesis equipment should be available as a precautionary measure. The patient's medical history for hypersensitivity reactions should be carefully evaluated prior to performing the intravitreal procedure (see section 4.3). Adequate anaesthesia and a broad-spectrum

topical microbicide to disinfect the periocular skin, eyelid and ocular surface should be administered prior to the injection.

The injection needle should be inserted 3.5 to 4.0 mm posterior to the limbus into the vitreous cavity, avoiding the horizontal meridian and aiming towards the centre of the globe. The injection volume of 0.05 ml is then delivered slowly; a different scleral site should be used for subsequent injections.

Immediately following the intravitreal injection, patients should be monitored for elevation in intraocular pressure. Appropriate monitoring may consist of a check for perfusion of the optic nerve head or tonometry. If required, sterile equipment for paracentesis should be available.

Following intravitreal injection patients should be instructed to report any symptoms suggestive of endophthalmitis (e.g. eye pain, redness of the eye, photophobia, blurring of vision) without delay.

Vial

The vial is for single use only. Each vial should only be used for the treatment of a single eye.

Since the volume contained in the vial (0.23 ml) is greater than the recommended dose (0.05 ml), a portion of the volume contained in the vial must be discarded prior to administration.

Injecting the entire volume of the vial could result in overdose. To expel the air bubble along with excess medicinal product, the air should be carefully expelled from the syringe and the dose adjusted to the 0.05 ml mark (equivalent to 50 µl, i.e. 6 mg brolocizumab).

For instructions on preparation of the medicinal product before administration, see section 6.6.

4.3 Contraindications

Hypersensitivity to the active substance or to any of the excipients listed in section 6.1.

Patients with active or suspected ocular or periocular infections.

Patients with active intraocular inflammation.

4.4 Special warnings and precautions for use

Traceability

In order to improve the traceability of biological medicinal products, the name and the batch number of the administered product should be clearly recorded.

Endophthalmitis, intraocular inflammation, traumatic cataract, retinal detachment, retinal tear, retinal vasculitis, and/or retinal vascular occlusion

Intravitreal injections, including those with Pagenax, have been associated with endophthalmitis, intraocular inflammation, traumatic cataract, retinal detachment and retinal tear (see section 4.8). Proper aseptic injection techniques must always be used when administering Pagenax.

Patients should be instructed to report any symptoms suggestive of the above-mentioned events without delay.

Intraocular inflammation, including retinal vasculitis and/or retinal vascular occlusion

Intraocular inflammation, including retinal vasculitis and/or retinal vascular occlusion, has been reported with the use of Pagenax (see sections 4.3 and 4.8). A higher number of intraocular inflammation events were observed among patients with treatment-emergent antibodies. After investigation, retinal vasculitis and/or retinal vascular occlusion were found to be immune-mediated events. Intraocular inflammation, including retinal vasculitis and/or retinal vascular occlusion, may occur following the first intravitreal injection and at any time of treatment. These events were observed more frequently at the beginning of the treatment.

Based on clinical studies these events were more frequent in female patients treated with Pagenax than male patients (e.g. 5.3% females vs. 3.2% males in HAWK and HARRIER) and in Japanese patients.

In patients developing these events, treatment with Pagenax should be discontinued and the events should be promptly managed. Patients treated with Pagenax with a medical history of intraocular inflammation and/or retinal vascular occlusion (within 12 months prior to the first brolocizumab injection) should be closely monitored, since they are at increased risk of developing retinal vasculitis and/or retinal vascular occlusion.

The interval between two Pagenax doses during maintenance treatment should not be less than 8 weeks considering that a higher incidence of intraocular inflammation (including retinal vasculitis) and retinal vascular occlusion was reported in patients with nAMD who received Pagenax every 4 week maintenance dosing in a clinical study compared to patients who received Pagenax every 8 or 12 week maintenance dosing in the pivotal Phase III clinical studies.

Intraocular pressure increases

Transient increases in intraocular pressure have been seen within 30 minutes of intravitreal injection with vascular endothelial growth factor (VEGF) inhibitors, including brolocizumab (see section 4.8). Special precaution is needed in patients with poorly controlled glaucoma (do not inject Pagenax while the intraocular pressure is ≥ 30 mmHg). Both intraocular pressure and perfusion of the optic nerve head must be monitored and managed appropriately.

Bilateral treatment

The safety and efficacy of brolocizumab administered in both eyes concurrently have not been studied.

Immunogenicity

As this is a therapeutic protein, there is a potential for immunogenicity with brolocizumab (see section 4.8). Patients should be instructed to inform their physician if they develop symptoms such as eye pain or increased discomfort, worsening eye redness, blurred or decreased vision, an increased number of small particles in their vision, or increased sensitivity to light (see section 4.8).

Concomitant use of other anti-VEGF

There are no data available on the concomitant use of Pagenax with other anti-VEGF medicinal products in the same eye. Brolocizumab should not be administered concurrently with other anti-VEGF medicinal products (systemic or ocular) (see section 4.5).

Withholding treatment

In intravitreal anti-VEGF treatments, the dose should be withheld and treatment should not be resumed earlier than the next scheduled treatment in the event of:

- a decrease in best-corrected visual acuity (BCVA) of ≥ 30 letters compared with the last assessment of visual acuity;
- a retinal break;
- a subretinal haemorrhage involving the centre of the fovea, or, if the size of the haemorrhage is $\geq 50\%$ of the total lesion area;
- performed or planned intraocular surgery within the previous or next 28 days.

Retinal pigment epithelial tear

Risk factors associated with the development of a retinal pigment epithelial tear after anti-VEGF therapy for wet AMD include a large and/or high pigment epithelial retinal detachment. When initiating brodalumab therapy, caution should be used in patients with these risk factors for retinal pigment epithelial tears.

Rhegmatogenous retinal detachment or macular holes

Treatment should be discontinued in subjects with rhegmatogenous retinal detachment or stage 3 or 4 macular holes.

Systemic effects following intravitreal use

Systemic adverse events, including non-ocular haemorrhages and arterial thromboembolic events, have been reported following intravitreal injection of VEGF inhibitors and there is a theoretical risk that these may relate to VEGF inhibition. There are limited data on safety in the treatment of patients with AMD and DME with a history of stroke, transient ischaemic attacks or myocardial infarction within the last 3 months. Caution should be exercised when treating such patients.

Populations with limited data

There is limited experience with Pagenax treatment in diabetic patients with HbA1c greater than 10% or with proliferative diabetic retinopathy. There is also no experience of treatment with Pagenax in diabetic patients with uncontrolled hypertension. This lack of information should be considered by the physician when treating such patients.

Sodium content

This medicinal product contains less than 1 mmol sodium (23 mg) per dose, that is to say essentially “sodium-free”.

Polysorbate 80 content

This medicinal product contains 0.01 mg polysorbate 80 per dose (0.05 ml). Polysorbates may cause allergic reactions. Patients need to be instructed to tell their doctor if they have any known allergies.

4.5 Interaction with other medicinal products and other forms of interaction

No interaction studies have been performed.

4.6 Fertility, pregnancy and lactation

Women of childbearing potential

Women of childbearing potential should use effective contraception during treatment with brolucizumab and for at least one month after the last dose when stopping treatment with brolucizumab.

Pregnancy

There are no or limited amount of data from the use of brolucizumab in pregnant women. A study in pregnant cynomolgus monkeys did not indicate any harmful effects with respect to reproductive toxicity. Animal studies are insufficient with respect to reproductive toxicity (see section 5.3). Although the systemic exposure after ocular administration is very low due to its mechanism of action, there is a potential risk to embryofoetal development. Therefore, brolucizumab should not be used during pregnancy unless the potential benefit outweighs the potential risk to the foetus.

Breast-feeding

It is unknown whether brolucizumab is excreted in human milk. In a reproductive toxicity study, brolucizumab was not detected in the maternal milk or infant serum of cynomolgus monkeys (see section 5.3). A risk to the breast-fed newborn/infant cannot be excluded. Brolucizumab is not recommended during breast-feeding and breast-feeding should not be started for at least one month after the last dose when stopping treatment with brolucizumab. A decision must be made whether to discontinue breast-feeding or to abstain from brolucizumab therapy, taking into account the benefit of breast-feeding for the child and the benefit of therapy for the woman.

Fertility

No reproductive or fertility studies have been conducted. VEGF inhibition has been shown to affect follicular development, corpus luteum function and fertility. Based on the mechanism of action of VEGF inhibitors, there is a potential risk for female reproduction.

4.7 Effects on ability to drive and use machines

Pagenax has a minor influence on the ability to drive and use machines due to possible temporary visual disturbances following the intravitreal injection and the associated eye examination. Patients should not drive or use machines until visual function has recovered sufficiently.

4.8 Undesirable effects

Summary of the safety profile

Wet AMD

For wet AMD, a total of 1 088 patients treated with brolucizumab constituted the safety population in two Phase III studies. Of these, 730 patients were treated with the recommended dose of 6 mg.

The most frequently reported adverse reactions were reduced visual acuity (7.3%), cataract (7.0%), conjunctival haemorrhage (6.3%) and vitreous floaters (5.1%).

The most serious adverse reactions were blindness (0.8%), endophthalmitis (0.7%), retinal artery occlusion (0.8%) and retinal detachment (0.7%).

DME

For DME, a total of 558 patients treated with brolucizumab constituted the safety population in two Phase III studies. Of these, 368 patients were treated with the recommended dose of 6 mg.

The most frequently reported adverse reactions were cataract (9.0%), conjunctival haemorrhage (6.5%) and intraocular pressure increased (5.4%).

The most serious adverse reactions were cataract (9.0%), retinal vascular occlusion (1.1%), retinal artery occlusion (0.8%), and endophthalmitis (0.5%).

Tabulated list of adverse reactions

The adverse reactions experienced following administration of Pagenax in clinical studies are summarised in Table 1 below.

Adverse reactions (Table 1) are listed according to the MedDRA system organ class. Within each system organ class, the adverse reactions are ranked by frequency, with the most frequent reactions first. Frequency categories for each adverse reaction are based on the following convention: very common ($\geq 1/10$), common ($\geq 1/100$ to $< 1/10$), uncommon ($\geq 1/1\ 000$ to $< 1/100$), rare ($\geq 1/10\ 000$ to $< 1/1\ 000$), very rare ($< 1/10\ 000$), not known (cannot be estimated from the available data). Within each frequency grouping, adverse reactions are presented in order of decreasing seriousness.

Table 1 **Frequencies of adverse reactions in clinical studies**

MedDRA System organ class	Frequency category*
Immune system disorders	
Hypersensitivity (including urticaria, rash, pruritus, erythema)	Common
Eye disorders	
Visual acuity reduced	Common
Retinal haemorrhage	Common
Uveitis	Common
Iridocyclitis	Common
Iritis	Common
Retinal vascular occlusion	Common
Vitreous haemorrhage	Common
Vitreous detachment	Common
Retinal tear	Common
Cataract	Common
Conjunctival haemorrhage	Common
Vitreous floaters	Common
Eye pain	Common
Intraocular pressure increase	Common
Conjunctivitis	Common
Retinal pigment epithelial tear	Common
Vision blurred	Common
Corneal abrasion	Common
Punctate keratitis	Common
Blindness	Uncommon
Endophthalmitis	Uncommon
Retinal detachment	Uncommon
Conjunctival hyperaemia	Uncommon
Lacrimation increased	Uncommon
Abnormal sensation in eye	Uncommon

Detachment of retinal pigment epithelium	Uncommon
Vitritis	Uncommon
Anterior chamber inflammation	Uncommon
Anterior chamber flare	Uncommon
Corneal oedema	Uncommon
Retinal vasculitis	Uncommon
Scleritis**	Uncommon
*The frequency category for each adverse reaction is based on the most conservative incidence rate from either pooled nAMD or pooled DME Phase III pivotal studies. **including episcleritis	

Description of selected adverse reactions

Immunogenicity

There is a potential for an immune response in patients treated with Pagenax.

Wet AMD

After dosing with Pagenax for 88 weeks, treatment-emergent anti-brolucizumab antibodies were detected in 23–25% of patients.

DME

After dosing with Pagenax for 96 weeks, treatment-emergent anti-brolucizumab antibodies were detected in 16-23% of patients.

Among AMD and DME patients with treatment-emergent antibodies, a higher number of intraocular inflammation adverse reactions were observed. After investigation, retinal vasculitis and/or retinal vascular occlusion, typically in the presence of intraocular inflammation, were found to be immune-mediated adverse events related to exposure to Pagenax(see section 4.4). Anti-brolucizumab antibodies were not associated with an impact on clinical efficacy.

Product-class-related adverse reactions

There is a theoretical risk of arterial thromboembolic events, including stroke and myocardial infarction, following intravitreal use of VEGF inhibitors. A low incidence rate of arterial thromboembolic events was observed in the brolucizumab clinical studies in patients with AMD and DME. There were no major notable differences between the groups treated with brolucizumab and comparator.

4.9 Overdose

Overdosing with greater than recommended injection volume may increase intraocular pressure. In the event of overdose, intraocular pressure should therefore be monitored and, if deemed necessary by the treating physician, appropriate treatment should be initiated.

5. PHARMACOLOGICAL PROPERTIES

5.1 Pharmacodynamic properties

Pharmacotherapeutic group: Ophthalmologicals, antineovascularisation agents, ATC code: S01LA06

Mechanism of action

Brolucizumab is a humanised monoclonal single chain Fv (scFv) antibody fragment with a molecular weight of ~26 kDa.

Increased levels of signalling through the vascular endothelial growth factor A (VEGF-A) pathway are associated with pathological ocular angiogenesis and retinal oedema. Brolucizumab binds with high affinity to VEGF-A isoforms (e.g. VEGF₁₁₀, VEGF₁₂₁, and VEGF₁₆₅), thereby preventing binding of VEGF-A to its receptors VEGFR-1 and VEGFR-2. By inhibiting VEGF-A binding, brolucizumab suppresses endothelial cell proliferation, thereby reducing pathological neovascularisation and decreasing vascular permeability.

Pharmacodynamic effects

Wet AMD

In the HAWK and HARRIER studies, anatomical parameters related to leakage of blood and fluid that characterise choroidal neovascularisation (CNV) were part of the disease activity assessments guiding treatment decisions. Reductions in central subfield thickness (CST) and in presence of intraretinal/subretinal fluid (IRF/SRF) or sub-retinal pigment epithelium (sub-RPE) fluid were observed in patients treated with Pagenax as early as 4 weeks after treatment initiation and up to week 48 and week 96.

At week 16, the reduction in CST was statistically significant on Pagenax versus aflibercept in both studies (HAWK: -161 vs. -134 microns; HARRIER: -174 vs. -134 microns). This decrease from baseline in CST was also statistically significant at week 48 (HAWK: -173 vs. -144 microns; HARRIER: -194 vs. -144 microns), and maintained to the end of each study at week 96 (HAWK: -175 vs. -149 microns; HARRIER: -198 vs. -155 microns).

At week 16, the percentage difference in patients with IRF and/or SRF fluid was statistically significant on Pagenax versus aflibercept in both studies (HAWK: 34% vs. 52%; HARRIER: 29% vs. 45%). This difference was also statistically significant at week 48 (HAWK: 31% vs. 45%; HARRIER: 26% vs. 44%), and maintained to the end of each study at week 96 (HAWK: 24% vs. 37%; HARRIER: 24% vs. 39%).

At week 16, the percentage difference in patients with sub-RPE fluid was statistically significant on Pagenax versus aflibercept in both studies (HAWK: 19% vs. 27%; HARRIER: 16% vs. 24%). This difference was also statistically significant at week 48 (HAWK: 14% vs. 22%; HARRIER: 13% vs. 22%), and maintained to the end of each study at week 96 (HAWK: 11% vs. 15%; HARRIER: 17% vs. 22%).

In these studies, for patients treated with Pagenax, reductions in CNV lesion size were observed as early as 12 weeks, and at weeks 48 and 96 after treatment initiation.

DME

In the KESTREL and KITE studies, related anatomical parameters were part of the disease activity assessments guiding treatment decisions. Reductions in CST and in presence of IRF/SRF were observed in patients treated with Pagenax as early as 4 weeks after treatment initiation and up to week 52. These reductions were maintained up to week 100.

Clinical efficacy and safety

Wet AMD

The efficacy and safety of Pagenax were assessed in two randomised, multicentre, double-masked, active-controlled Phase III studies (HAWK and HARRIER) in patients with neovascular (wet) AMD. A total of 1,817 patients were treated in these studies for two years (1 088 on Pagenax and 729 on comparator aflibercept). Patient ages ranged from 50 to 97 years, with a mean age of 76 years.

In both studies, after the first three monthly doses (weeks 0, 4 and 8), brolucizumab patients were treated every 12 weeks, with the option of adjusting to a dosing interval every 8 weeks based on

disease activity. Disease activity was assessed by a physician during the first 12-week interval (at weeks 16 and 20) and at each subsequent scheduled 12-weekly treatment visit. Patients who showed disease activity (e.g. decreased visual acuity, increased CST and/or presence of IRF/SRF or sub-RPE fluid) at any of these visits were adjusted to an 8-weekly treatment interval. The comparator aflibercept was administered every 8 weeks after the first 3 monthly doses.

Results

The primary efficacy endpoint for the studies was the change from baseline in best corrected visual acuity (BCVA) to week 48, as measured by the early treatment diabetic retinopathy study (ETDRS) letter score, with the primary objective being to demonstrate non-inferiority of Pagenax versus aflibercept. In both studies, Pagenax (administered in an every 12 weeks or an every 8 weeks regimen) demonstrated non-inferior efficacy to aflibercept 2 mg (administered every 8 weeks). The visual acuity gains observed in the first year were maintained in the second year.

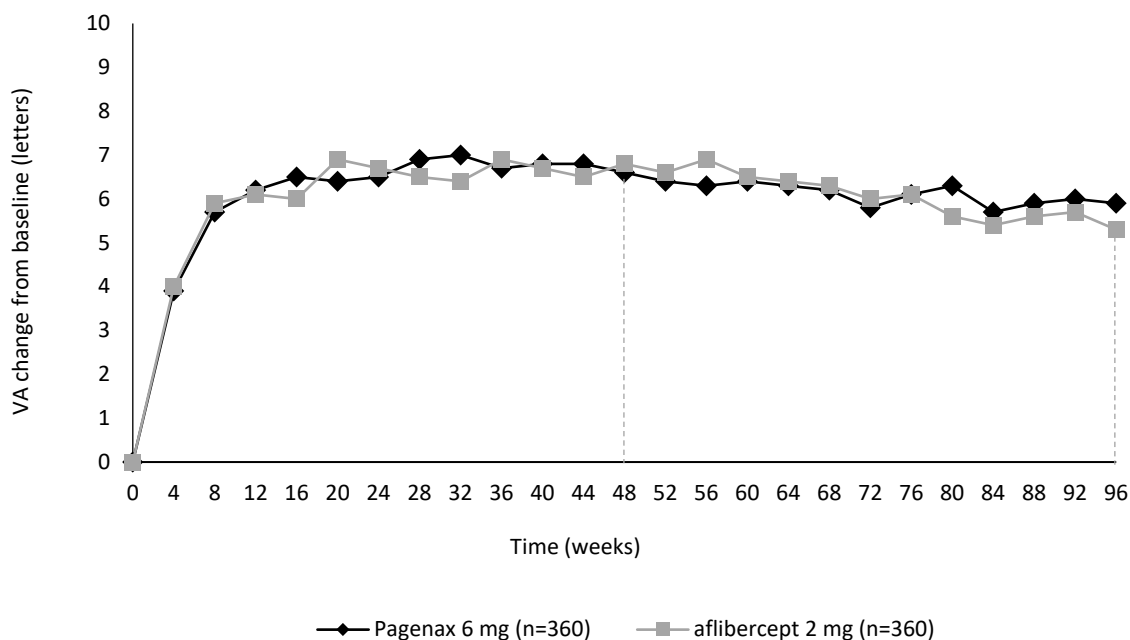
Detailed results of both studies are shown in Table 2 and in Figure 1 below.

Table 2 Visual acuity outcomes at weeks 48 and 96 in Phase III - HAWK and HARRIER studies

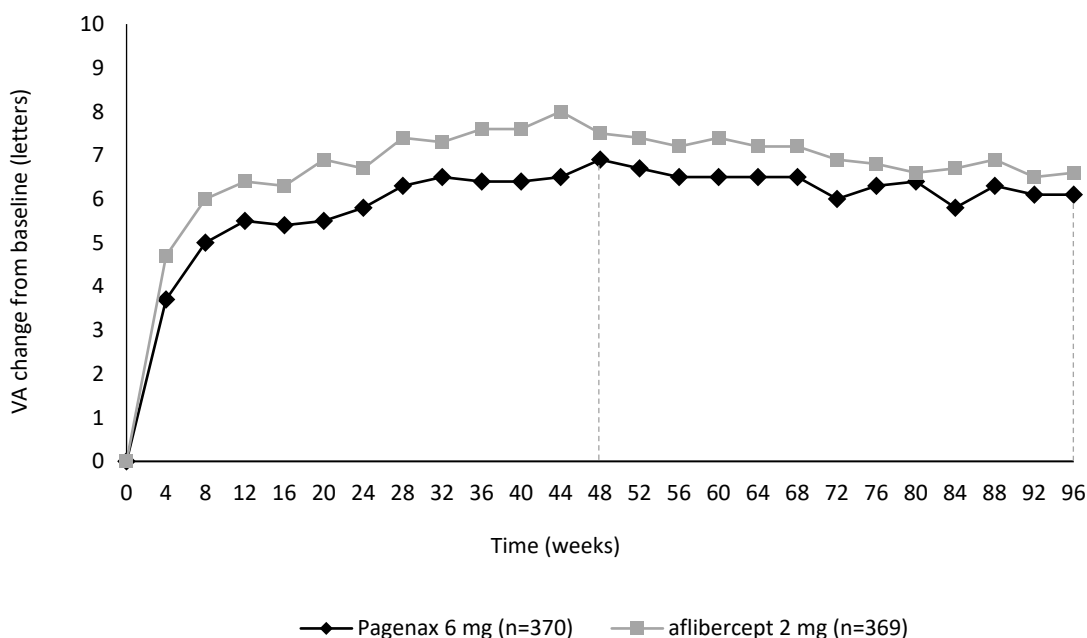
Efficacy outcome	Week	HAWK			HARRIER		
		Pagenax 6 mg (n=360)	Aflibercept 2 mg (n=360)	Difference (95% CI) brolucizumab – aflibercept	Pagenax 6 mg (n=370)	Aflibercept 2 mg (n=369)	Difference (95% CI) brolucizumab – aflibercept
Mean change from baseline in BCVA (measured by ETDRS letters score)	48	6.6 (SE=0.71)	6.8 (SE=0.71)	-0.2 (-2.1, 1.8) P<0.0001 ^{a)}	6.9 (SE=0.61)	7.6 (SE=0.61)	-0.7 (-2.4, 1.0) P<0.0001 ^{a)}
	36 – 48 ^{b)}	6.7 (SE=0.68)	6.7 (SE=0.68)	0.0 (-1.9, 1.9) P<0.0001 ^{a)}	6.5 (SE=0.58)	7.7 (SE=0.58)	-1.2 (-2.8, 0.4) P=0.0003 ^{a)}
	96	5.9 (SE=0.78)	5.3 (SE=0.78)	0.5 (-1.6, 2.7)	6.1 (SE=0.73)	6.6 (SE=0.73)	-0.4 (-2.5, 1.6)
% of patients who gained at least 15 letters of vision	48	33.6	25.4	8.2 (2.2, 15.0)	29.3	29.9	-0.6 (-7.1, 5.8)
	96	34.2	27.0	7.2 (1.4, 13.8)	29.1	31.5	-2.4 (-8.8, 4.1)
% of patients who lost visual acuity (%) (≥15 letters of BCVA loss)	48	6.4	5.5	0.9 (-2.7, 4.3)	3.8	4.8	-1.0 (-3.9, 2.2)
	96	8.1	7.4	0.7 (-3.6, 4.6)	7.1	7.5	-0.4 (-3.8, 3.3)
BCVA: best corrected visual acuity; missing data are imputed using last observation carried forward (LOCF) method ETDRS: early treatment diabetic retinopathy study SE: standard error ^{a)} P-value referring to the non-inferiority hypothesis with a non-inferiority margin of 4.0 letters. ^{b)} Key secondary endpoint, accounting for differences in timing of Pagenax and aflibercept treatments.							

Figure 1 Mean change in visual acuity from baseline to week 96 in HAWK and HARRIER studies

HAWK



HARRIER



These visual acuity gains were achieved with 56% and 51% of patients treated with Pagenax 6 mg on a 12-weekly dosing interval at week 48, and with 45% and 39% of patients at week 96 in HAWK and HARRIER, respectively. Among patients identified as eligible for the 12-weekly regimen during the first 12-week interval, 85% and 82% remained on the 12-weekly dosing interval up to week 48. Of patients on the 12-weekly interval at week 48, 82% and 75% remained on the 12-weekly dosing interval up to week 96.

Treatment effects in evaluable subgroups (e.g. age, gender, race, baseline visual acuity, baseline retinal thickness, lesion type, lesion size, fluid status) in each study were generally consistent with the results in the overall populations.

Disease activity was assessed by changes in visual acuity and/or anatomical parameters, including CST and/or presence of IRF/SRF or sub-RPE. Disease activity was assessed throughout the studies. Anatomical parameters of disease activity were decreased at week 48 and at week 96 for Pagenax compared to aflibercept (see “Pharmacodynamic effects”).

The percentage difference in patients with disease activity at week 16 was statistically significant on Pagenax versus aflibercept (24% vs 35% in HAWK, $p=0.0013$; 23% vs 32% in HARRIER, $p=0.0021$).

In both studies, Pagenax demonstrated clinically meaningful increases from baseline in the pre-specified secondary efficacy endpoint of patient-reported outcomes, reported through the National Eye Institute Visual Function Questionnaire (NEI VFQ-25). The magnitude of these changes was similar to that seen in published studies, which corresponded to a 15-letter gain in BCVA. Patient-reported outcome benefits were maintained in the second year.

No clinically meaningful differences were found between Pagenax and aflibercept in changes from baseline to week 48 in NEI VFQ-25 total score and subscales (general vision, ocular pain, near activities, distance activities, social functioning, mental health, role difficulties, dependency, driving, colour vision and peripheral vision).

The results of the Pagenax arms of the HAWK and HARRIER studies, where Pagenax was administered every 4 weeks (monthly) for the first 3 doses (loading) followed by maintenance dosing every 12 or 8 weeks, were replicated in a population pharmacokinetic/pharmacodynamic model simulation study where Pagenax was administered every 6 weeks for the first 2 or 3 doses (loading) followed by maintenance dosing every 12 or 8 weeks.

A treat-and-extend dosing regimen for the maintenance phase was examined in the TALON study, which was a 64-week, two-arm, randomised, double-masked, multicentre, Phase IIIb study assessing the efficacy and safety of Pagenax compared to aflibercept 2 mg in patients with nAMD.

737 patients were randomised in a 1:1 ratio to one of the two treatment arms, either brolucizumab 6 mg or aflibercept 2 mg. Patients in both treatment arms were dosed once every 4 weeks for the first 3 injections and then one injection after 8 weeks. Thereafter, treatment intervals were either every 8 weeks, every 12 weeks, or every 16 weeks up to week 60 or 62.

The average change in BCVA from baseline at week 64 was +4.7 ETDRS letters vs. +4.9 ETDRS letters for Pagenax and aflibercept 2 mg, respectively.

Results of treatment intervals at week 64 are presented in Table 3.

Table 3 Last treatment interval with no disease activity: proportion of patients at week 64

Interval (weeks)	Study arm	
	Brolucizumab 6 mg n=366	Aflibercept 2 mg n=368
4	23.2%	41.8%
8	26.0%	22.0%
12	22.4%	23.9%
16	28.4%	12.2%

255 subjects who completed the TALON study were enrolled into a 56-week open-label, one-arm extension study of TALON and treated with a brolucizumab treat-and-extend dosing regimen without a loading phase and with a maximum treatment interval of up to 20 weeks.

At week 56, more than 50% of 237 subjects who had received at least 2 injections were on a treatment interval of 16 weeks (24.9%) or 20 weeks (28.7%) and had no disease activity, while visual acuity was maintained throughout the study.

DME

The efficacy and safety of Pagenax were assessed in two randomised, multicentre, double-masked, active-controlled Phase III studies (KESTREL and KITE) in patients with visual impairment due to diabetic macular oedema. A total of 926 patients were treated in these studies for two years (558 on brolucizumab and 368 on aflibercept 2 mg). Patient ages ranged from 23 to 87 years, with a mean age of 63 years.

In both studies, after the first five doses (weeks 0, 6, 12, 18 and 24), brolucizumab patients were treated every 12 weeks, with the option of adjusting to a dosing interval every 8 weeks based on disease activity. Disease activity was assessed by a physician during the first 12-week interval (at weeks 32 and 36) and at each subsequent scheduled treatment visit. Patients who showed disease activity (e.g. decreased visual acuity, increased CST) at any of these visits were adjusted to an every 8 weeks treatment interval. In year 2 of KITE, patients who showed no disease activity could be extended to a 16-week treatment interval. The comparator aflibercept was administered every 8 weeks after the first 5 monthly doses.

Results

The primary efficacy endpoint for the studies was the change from baseline in BCVA to week 52, as measured by the ETDRS letter score, with the primary objective being to demonstrate non-inferiority of Pagenax versus aflibercept 2 mg. In both studies, Pagenax (administered in an every 12 weeks or an every 8 weeks regimen) demonstrated non-inferior efficacy to aflibercept 2 mg (administered every 8 weeks).

The results of KESTREL and KITE also demonstrated non-inferiority of Pagenax versus aflibercept 2 mg for the key secondary endpoint (average change from baseline in BVCA over the period week 40 to week 52).

The visual acuity gains observed in the first year were maintained in the second year.

Detailed results of both studies are shown in Table 4 and in Figure 2 below.

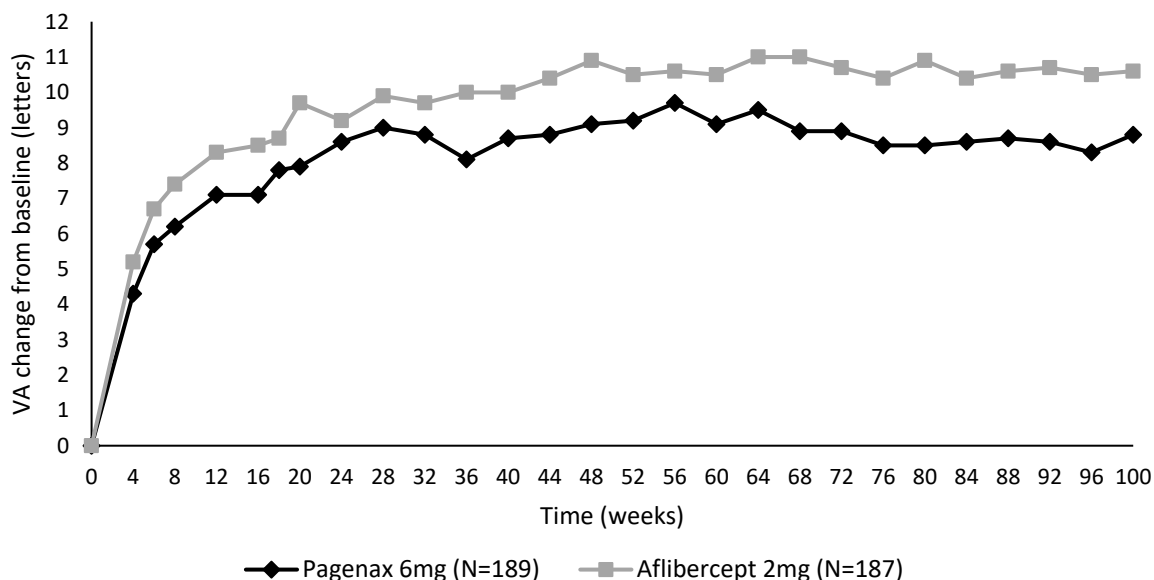
Table 4 Visual acuity outcomes at weeks 52 and 100 in Phase III - KESTREL and KITE studies

Efficacy outcome	Week	KESTREL			KITE		
		Pagenax (n=189)	Aflibercept 2 mg (n=187)	Difference (95% CI) brolucizumab – aflibercept	Pagenax (n=179)	Aflibercept 2 mg (n=181)	Difference (95% CI) brolucizumab– aflibercept
Change from baseline in BCVA (measured by ETDRS letters score) – LS mean (SE)	52	9.2 (0.57)	10.5 (0.57)	-1.3 (-2.9, 0.3) P <0.001 ^a	10.6 (0.66)	9.4 (0.66)	1.2 (-0.6, 3.1) P <0.001 ^a
	40-52	9.0 (0.53)	10.5 (0.53)	-1.5 (-3.0, 0.0) P <0.001 ^a	10.3 (0.62)	9.4 (0.62)	0.9 (-0.9, 2.6) P <0.001 ^a
	100	8.8 (0.75)	10.6 (0.75)	-1.7 (-3.8, 0.4)	10.9 (0.85)	8.4 (0.85)	2.6 (0.2, 4.9)

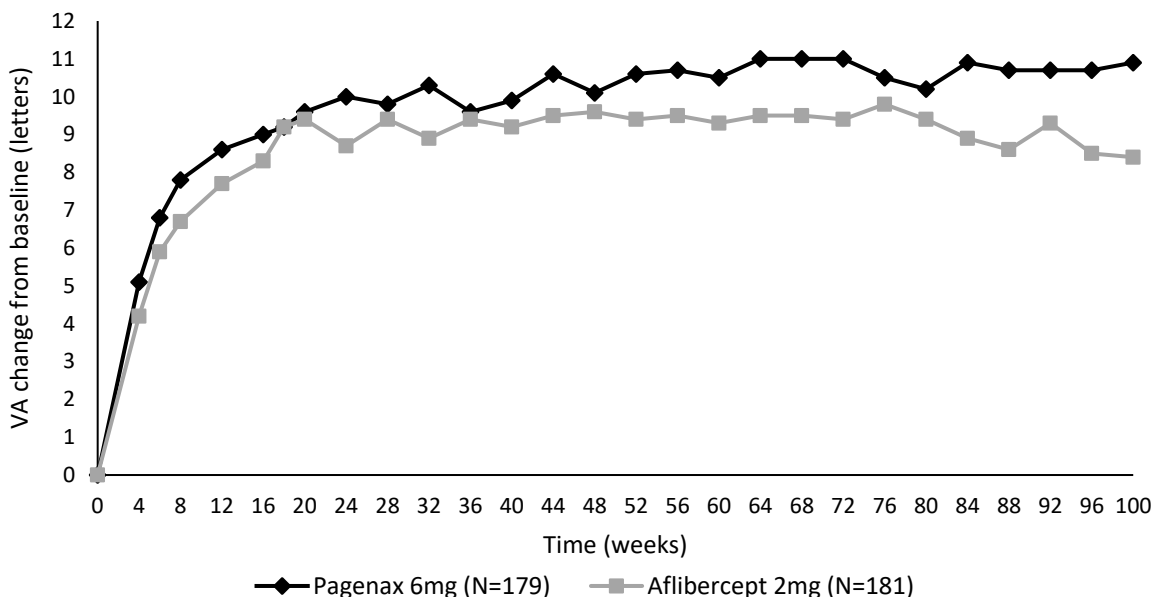
Gain of at least 15 letters in BCVA from baseline or BCVA $\geq$84 letters (%)	52	36.0	40.1	-4.1 (-13.3, 5.9)	46.8	37.2	9.6 (-0.4, 20.2)
	100	39.2	42.2	-3.0 (-12.5, 6.3)	50.4	36.9	13.6 (3.3, 23.5)
BCVA: best corrected visual acuity; BCVA assessments after start of alternative DME treatment in the study eye were censored and replaced by the last value prior to start of this alternative treatment. ETDRS: early treatment diabetic retinopathy study LS: least-square SE: standard error ^a P-value referring to the non-inferiority hypothesis with a non-inferiority margin of 4.0 letters							

Figure 2 Mean change in visual acuity from baseline to week 100 in KESTREL and KITE studies

KESTREL



KITE



These visual acuity gains were achieved with 55% and 50% of patients treated with Pagenax on a 12-weekly dosing interval at week 52, and 44% and 37% of patients treated with Pagenax on a 12-weekly or 12-weekly/16-weekly dosing interval at week 100 in KESTREL and KITE, respectively. Among patients identified as eligible for the 12-weekly regimen during the first 12-week interval, approximately 70% remained on at least the 12-weekly interval at week 100 in both studies. In KITE, 25% of patients were treated with Pagenax on a 16-weekly dosing interval at week 100.

Treatment effects in evaluable subgroups (e.g. age, gender, baseline HbA1c, baseline visual acuity, baseline central subfield thickness, DME lesion type, duration of DME since diagnosis, retinal fluid status) in each study were generally consistent with the results in the overall populations.

In KESTREL and KITE, disease activity was assessed throughout the studies by changes in visual acuity and/or anatomical parameters, including CST and/or presence of IRF/SRF. The reduction in CST from baseline was maintained up to week 100. At week 100, the proportion of patients with IRF/SRF was lower in patients treated with Pagenax (42% KESTREL and 41% KITE) compared to patients treated with aflibercept 2 mg (54% KESTREL and 57% KITE).

Diabetic retinopathy severity score (DRSS) was assessed in the KESTREL and KITE studies. At baseline, 98.1% of patients in both KESTREL and KITE had gradable DRSS scores. Based on the pooled analysis, Pagenax showed non-inferiority to aflibercept 2 mg in the proportion of subjects with at least a 2-step improvement from baseline in DRSS at week 52, using a non-inferiority margin of 10%. Estimated proportions were 28.9% and 24.9% in Pagenax and aflibercept 2 mg, respectively, resulting in a treatment difference of 4.0% (95% CI: [-0.6, 8.6]). At week 100, the proportion of patients with a ≥ 2 -step improvement from baseline to week 100 in the DRSS score was 32.8% with Pagenax and 29.3% with aflibercept 2 mg in KESTREL and 35.8% with Pagenax and 31.1% with aflibercept 2 mg in KITE.

5.2 Pharmacokinetic properties

Pagenax is administered directly into the vitreous to exert local effects in the eye.

Absorption and distribution

After intravitreal administration of 6 mg brolocizumab per eye to patients with nAMD, the geometric mean C_{max} of free brolocizumab in the plasma was 49.0 ng/ml (range: 8.97 to 548 ng/ml) and was attained in 1 day.

Biotransformation and elimination

Brolocizumab is a monoclonal antibody fragment and no metabolism studies have been conducted. As a single-chain antibody fragment, free brolocizumab is expected to undergo elimination through both target-mediated disposition via binding to free endogenous VEGF, passive renal elimination and metabolism via proteolysis.

After intravitreal injections, brolocizumab was eliminated with an apparent systemic half-life of 4.3 ± 1.9 days. Concentrations were generally near or below the quantitation limit (<0.5 ng/ml) approximately 4 weeks after dosing in most patients. Brolocizumab did not accumulate in the serum when administered intravitreally every 4 weeks.

Special populations

Elderly

There were no relevant differences in systemic pharmacokinetics following intravitreal injection in a study with 22 patients aged 65 to 74 years, 18 patients aged 75 to 84 years and 3 patients aged ≥ 85 years.

Renal impairment

The systemic pharmacokinetics of brolocizumab was evaluated in nAMD patients with normal renal function (≥ 90 ml/min [$n=21$]), with mild (60 to <90 ml/min [$n=22$]) or moderate (30 to <60 ml/min [$n=7$]) renal impairment. While the mean systemic clearance values for patients with mild or moderate renal impairment were generally lower than patients with normal renal function, no significant impact

of mild and moderate renal impairment on the overall systemic exposure to brolocizumab was observed. No patients with severe (<30 ml/min) renal impairment were studied.

Hepatic impairment

Brolocizumab has not been studied in patients with hepatic impairment. Mild to severe hepatic impairment should have no impact on the overall systemic exposure to brolocizumab, because metabolism occurs via proteolysis and does not depend on hepatic function.

5.3 Preclinical safety data

No studies have been conducted on the carcinogenic or mutagenic potential of brolocizumab.

In pregnant cynomolgus monkeys, brolocizumab was administered once every 4 weeks by intravitreal injection at dose levels resulting in maximal systemic exposures 6-fold higher than those in humans at the maximum recommended dose (based on serum C_{max}). There was no impact on embryofoetal development, pregnancy or parturition, or on the survival, growth or postnatal development of offspring. Nevertheless, based on its pharmacological effect, brolocizumab should be regarded as potentially teratogenic and embryo-foetotoxic.

6. PHARMACEUTICAL PARTICULARS

6.1 List of excipients

Sodium citrate
Sucrose
Polysorbate 80
Water for injections

6.2 Incompatibilities

In the absence of compatibility studies, this medicinal product must not be mixed with other medicinal products.

6.3 Special precautions for storage

Vial

Store in a refrigerator at 2°C to 8°C.
Do not freeze.
Keep the vial in the outer carton in order to protect from light.

6.5 Nature and contents of container

Vial

0.230 ml sterile solution in a glass vial with a coated rubber stopper sealed with an aluminium cap with a purple plastic flip-off disk.

Pack size of 1 vial and 1 blunt filter needle (18G x 1½", 1.2 mm x 40 mm, 5 µm).

6.6 Special precautions for disposal and other handling

Vial

The vial contains more than the recommended dose of 6 mg. The extractable volume of the vial (0.23 ml) is not to be used in total. The excess volume should be expelled prior to injection. Injecting the entire volume of the vial could result in overdose. The injection dose must be set to the 0.05 ml dose mark, i.e. 6 mg brolocizumab.

The solution should be inspected visually upon removal from the refrigerator and prior to administration. If particulates or cloudiness are visible, the vial must not be used, and appropriate replacement procedures must be followed.

The content of the vial and the filter needle are sterile and for single use only. Do not use if the packaging, vial and/or filter needle are damaged or expired. Detailed instructions for use are provided in the package leaflet.

Any unused medicinal product or waste material should be disposed of in accordance with local requirements.

6.7 Instructions for use and handling

Instructions for use of the Pagenax vial kit

Storage and inspection



Store Pagenax in the refrigerator (2°C to 8°C/36°F to 46°F); do not freeze. Keep the vial in the outer carton to protect from light.



After opening the vial, proceed under aseptic conditions.



Pagenax is a clear to slightly opalescent and colorless to slightly brownish-yellow solution.



The solution should be inspected visually upon removal from the refrigerator and prior to administration. If particulates or cloudiness are visible, the vial must not be used and appropriate replacement procedures followed.

The contents of the vial and filter needle are sterile and for single use only. Do not use if the packaging, vial and/or filter needle are damaged or expired.

How to prepare and administer Pagenax

The intravitreal injection procedure must be carried out under aseptic conditions, which includes the use of surgical hand disinfection, sterile gloves, a sterile drape and a sterile eyelid speculum (or equivalent) and the availability of sterile paracentesis equipment (if required). Adequate anesthesia and a broad-spectrum topical microbicide to disinfect the periocular skin, eyelid and ocular surface should be administered prior to the injection.

For preparation and intravitreal injection the following single use medical devices are needed:

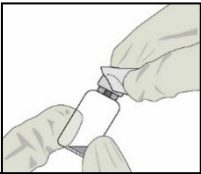
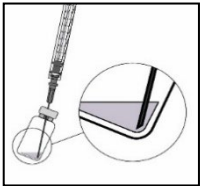
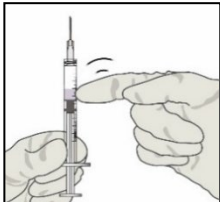
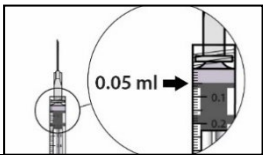
- A 30G x ½” injection needle, sterile.
- A 1 mL syringe with a 0.05 mL dose mark, sterile

The injection needle and the syringe are not included in the Pagenax vial kit.

Note: The dose must be set to 0.05 mL.

Ensure that the injection is given immediately after preparation of the dose (Step 8).

Injection procedure

1		Remove the vial cap and clean the vial septum (e.g. with 70% alcohol swab).
2		Assemble the filter needle onto a 1 mL syringe using aseptic technique.
3		Push the filter needle into the center of the vial septum until the needle touches the bottom of the vial.
4		To withdraw the liquid, hold the vial slightly inclined and slowly withdraw all the liquid from the vial and filter needle. Ensure that the plunger rod is drawn sufficiently back when emptying the vial in order to completely empty the filter needle.
5		Disconnect the filter needle from the syringe in an aseptic manner and dispose of it. The filter needle is not to be used for intravitreal injection.
6		Aseptically and firmly assemble a 30G x 1/2" injection needle onto the syringe.
7		To check for air bubbles, hold the syringe with the needle pointing up. If there are any air bubbles, gently tap the syringe with your finger until the bubbles rise to the top.
8		Carefully expel the air from the syringe and adjust the dose to the 0.05 mL mark. The syringe is ready for the injection.
9		Inject slowly until the rubber stopper reaches the end of the syringe to deliver the volume of 0.05 mL. Confirm delivery of the full dose by checking that the rubber stopper has reached the end of the syringe barrel. Note: Any unused medicinal product or waste material should be disposed of in accordance with local regulations.

Commonly asked questions and answers

Q: What if I have difficulty withdrawing sufficient liquid from the vial?

A: Do not shake the vial before withdrawal but let the liquid settle to the bottom of the vial. Ensure the vial is in an upright, slightly inclined position. **Slowly withdraw** the plunger and wait for the liquid to appear in the syringe barrel. Continue to withdraw slowly to completely empty the vial and the filter needle.

Q: What if I cannot remove all the air bubbles from the liquid?

A: It is important that the liquid is air free. However, tiny air bubbles that are attached to the stopper usually do not detach from the stopper during the injection and therefore do not affect the dose volume.

7. Product Registration Holder

Novartis Corporation (Malaysia) Sdn. Bhd.
Level 18, Imazium,
No. 8, Jalan SS21/37, Damansara Uptown,
47400 Petaling Jaya, Malaysia.

8. Malaysia Package Leaflet

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